

$$E \quad \Theta \quad \Gamma \Lambda \quad - \frac{30 \Theta \quad \Delta (M \quad 2)}{\quad}$$

ΣΗΜΕΙΩΣΗ: T 15  $\Theta \quad \Delta$ .

  $\Theta \quad \Delta 16$

$$\Delta \quad f(x) = \begin{cases} \frac{x \ln x}{x-1}, & 0 < x \neq 1 \\ 1, & x = 1 \end{cases}$$

$$\Delta 1 \text{ N} \quad f \quad x_0 = 1.$$

$$\Delta 2 \text{ N} \quad f \quad x_0 = 1 \quad f'(1).$$

$$\Delta 3 \text{ N} \quad 0 \quad .$$

$$\Delta 4 \text{ N} \quad f'(x) > 0 \quad x > 0, x \neq 1, \quad .$$

  $\Theta \quad \Delta 17$

$$E \quad f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(0) = 0$$

$$f''(x) - 2f'(x) + f(x) = 0 \quad f'(0) = 1$$

$$\Delta 1 \text{ N} \quad g(x) = f(x)e^{-x} \quad g''(x) = 0 \quad x \in \mathbb{R}.$$

$$\Delta 2 \text{ N} \quad f(x) \quad f(x) = xe^x.$$

$$\Delta 3 \text{ N} \quad f \quad , \quad , \quad .$$

$$\Delta 4 \text{ N} \quad f(x^2) - f(3x) < 0 \quad \mathbb{R}.$$

  $\Theta \quad \Delta 18$

$$\Theta \quad f(x) = (x-2)e^x + x + 2, \quad x \in \mathbb{R}.$$

$$\Delta 1 \text{ N} \quad , \quad ( \quad ) \quad .$$

$$\Delta 2 \text{ } \Sigma \quad C_f \quad ;$$

$$\Delta 3 \text{ N} \quad C_f \quad -\infty.$$

$$\Delta 4 \text{ N} \quad C_f, \quad -\infty, \quad x = 0, x = -1.$$

  $\Theta \quad \Delta 19$

$$E \quad f(x) = \ln(x^2 + 1) - x, \quad x \in \mathbb{R}.$$

$$\Delta 1 \text{ N} \quad f \quad \mathbb{R} \quad f(e^x - 1) = f(0).$$

- $\Delta 2$  N  $f$  .
- $\Delta 3$  N  $C_f$  ” ” .
- $\Delta 4$   $\Upsilon$   $\lim_{x \rightarrow +\infty} \frac{f(x)+x}{x^2}$  .

  $\Theta$   $\Delta 20$

- $\Delta$   $f(x) = \frac{e^x}{x^2+1} \quad x \in \mathbb{R}$ .
- $\Delta 1$  N  $f$  . ( $\Pi$  :  $x = 1$  ).
- $\Delta 2$  N  $\frac{e^x}{x^2+1} \geq \frac{e}{2} \quad x \in \mathbb{R}$ .
- $\Delta 3$  N  $f$  . ( $X$   $\pm\infty$ ).
- $\Delta 4$  N  $f(\ln x) - f(\frac{1}{x}) = 0$ .

  $\Theta$   $\Delta 21$

- $\Sigma$   $Q(t) \quad Q'(t) = -kQ(t)$   
 $f'(x) = f(x)e^x \quad f(0) = e$ .
- $\Delta 1$  N  $f(x) = e^{e^x}$  .
- $\Delta 2$  N  $f$  ( ).
- $\Delta 3$  N  $\lim_{x \rightarrow -\infty} f(x)$  .
- $\Delta 4$  N  $\ln f(x) \quad x = 0$ .

  $\Theta$   $\Delta 22$

- $\Theta$   $f(x) = 2x + \ln(x^2 + 1)$ .
- $\Delta 1$  N  $f$   $f^{-1}$ .
- $\Delta 2$  N  $f(x^2 - x) > f(x + 3)$ .
- $\Delta 3$  N  $f$  .
- $\Delta 4$  N  $C_f$ ,  $x'x$   $x = 0, x = 1$ .

✍ Θ Δ23

E  $f(x) = \left(x + \frac{1}{x}\right) \ln x, \quad x > 0.$

Δ1 N  $x = 0$   $C_f.$

Δ2 N  $f'(x) : \alpha. x^2 - 1 + 2 \ln x > 0 \quad x > 1. \beta. M \quad f.$

Δ3 N  $\left( \quad \right) \left(x + \frac{1}{x}\right) \ln x > 0.$

Δ4 N  $\int_1^e \frac{f(x)}{x} dx.$

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✍ Θ Δ24

Δ  $f(x) = e^x - ax, \quad \alpha > 0$  .

Δ1 N  $f$  .

Δ2 Γ  $a > 0,$  ;

Δ3 A  $a = e,$   $e^{x-1} \geq x \quad x \in \mathbb{R}.$

Δ4 X  $x = \int_0^1 f(t) dt,$  .

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✍ Θ Δ25

E  $f \quad e^{f(x)} + f(x) = x \quad x \in \mathbb{R}.$

Δ1 Δ  $1 - 1 \quad f^{-1}(y).$

Δ2 M  $f' \quad (X \quad ).$

Δ3 N  $\lim_{x \rightarrow +\infty} f(x),$  .

Δ4 N  $C_{f^{-1}}, \quad x \quad x = 1, x = e + 1.$

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✍ Θ Δ26

Δ  $I_v = \int_0^1 x^v e^x dx, \quad v \in \mathbb{N}^*.$

Δ1 N  $I_1 \quad I_2.$

Δ2 N  $: I_v = e - v I_{v-1}.$

Δ3 X  $, \quad I_v \quad (I_{v+1} < I_v).$

Δ4 N  $\lim_{v \rightarrow +\infty} I_v.$

✎ Θ Δ27

M  $f : [0, +\infty) \rightarrow \mathbb{R}$   $f(0) = 1,$  :

$$f'(x) = \frac{1}{1 + f^2(x)} \quad x \geq 0$$

Δ1 N  $f$  .

Δ2 O  $(1 + f^2(x))f'(x) = 1,$   $f(x)$   $x$ .

Δ3 A  $x > 0,$  :  $\frac{x}{1+f^2(x)} < f(x) - 1 < x$ .

Δ4 N  $f(x) + f^3(x) = x + 2$  (Δ2).

✎ Θ Δ28

Δ  $f(x) = \sqrt{x^2 - x + 1} - ax, \quad a \in \mathbb{R}.$

Δ1 A  $\lim_{x \rightarrow +\infty} f(x) = \frac{1}{2},$   $a.$  ( $a = 1$ ).

Δ2 Γ  $a = 1,$   $C_f$   $+\infty$ .

Δ3 Γ  $a = 1,$   $-\infty$ .

Δ4 Υ  $\int_0^1 \frac{2x-1}{2\sqrt{x^2-x+1}} dx.$

✎ Θ Δ29

E  $f(x) = x \sin x + \cos x, \quad [0, 2\pi].$

Δ1 B  $f'(x)$   $f$   $[0, 2\pi].$

Δ2 A  $f$   $[0, 2\pi].$

Δ3 Δ  $c \in (f(\pi), f(\frac{\pi}{2})),$   $f(x) = c$   $(0, 2\pi).$

Δ4 N  $C_f$   $C_{f'}$  .

✎ Θ Δ30

E Π P M & A

E  $M$   $y = \frac{1}{x}, \quad x > 0.$  H  $x$   $M$  2 cm/s  
( $x'(t) = 2$ ).

Δ1 N  $y(t)$   $t_0$   $M$   $(2, \frac{1}{2}).$

|            |          |              |          |     |                   |          |       |                 |
|------------|----------|--------------|----------|-----|-------------------|----------|-------|-----------------|
| $\Delta 2$ | $\Theta$ | $OAM$        | $O(0,0)$ | $A$ | $M$               | $x'x.$   | $\Pi$ | $OAM;$          |
| $\Delta 3$ | H        | $(\epsilon)$ | $M$      |     | $\Gamma, \Delta.$ | N        |       | $O\Gamma\Delta$ |
|            | (        | $t)$         | 2.       |     |                   |          |       |                 |
| $\Delta 4$ | N        |              | $(OM)$   |     | $x.$              | $\Sigma$ | $M$   | ;               |

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